

Chapter 2: Polynomial Regression and Transformations in R

Syntax Notes, Exercises, and Worked Solutions

STAT452 – Applied Statistics for Engineers and Scientists II

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1 What this worksheet covers

Polynomial regression models curvature by adding powers of a predictor. A model such as $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$ is nonlinear in X but linear in its coefficients, so it is fit with `lm()`. We compare transformations, centering, model order, and extrapolation.

2 Core R syntax

```
linear <- lm(mpg ~ wt, data=cars)
quadratic <- lm(mpg ~ wt + I(wt^2), data=cars)
anova(linear, quadratic)
AIC(linear, quadratic); BIC(linear, quadratic)
predict(quadratic, data.frame(wt=3))
```

Use `I(wt^2)` to calculate a numerical square inside a formula. Keep lower-order terms when higher-order terms are present (hierarchy). Centering uses $X_c = X - \bar{X}$ and often improves numerical conditioning.

3 Exercises

Exercise 1: Explore the relationship. Load `mtcars`, plot `mpg` against `wt`, and describe why a straight line may be inadequate.

Exercise 2: Choose polynomial order. Fit first-, second-, and third-order models. Compare nested models with `anova()`, AIC, and BIC. Which model is defensible and why?

Exercise 3: Interpret curvature. For the quadratic model, calculate $x^* = -\hat{\beta}_1 / (2\hat{\beta}_2)$. Explain why individual coefficients are less directly interpretable than the fitted curve.

Exercise 4: Centering and collinearity. Compare the condition numbers of uncentered and centered quadratic design matrices.

Exercise 5: Transform and diagnose. Fit $\log(Y)$ on $\log(X)$, inspect residual plots, and predict at in-range and out-of-range weights.

4 Worked solutions

Solution 1. The scatterplot is curved: fuel efficiency falls quickly at low-to-moderate weights and then flattens. A quadratic term is a reasonable first departure from linearity. $\square$

Solution 2. Use `anova(linear, quadratic, cubic)`. Retain an extra term only when the improvement justifies complexity. Select the simplest adequate model, not automatically the highest order. $\square$

Solution 3. Since the derivative is $\beta_1 + 2\beta_2x$, the stationary point is $-\hat{\beta}_1/(2\hat{\beta}_2)$. Interpret slopes locally or through predicted changes. $\square$

Solution 4. Powers of the same variable are correlated. Centering reduces non-essential ill-conditioning and improves numerical stability. $\square$

Solution 5. Log transformations can turn multiplicative relationships into additive ones and reduce skewness. A polynomial can behave wildly outside the observed predictor range, so extrapolation requires scientific justification. $\square$

Checklist

```
plot(y ~ x); fit1 <- lm(y ~ x, d); fit2 <- lm(y ~ x + I(x^2), d)
anova(fit1, fit2); AIC(fit1, fit2); BIC(fit1, fit2)
par(mfrow=c(2,2)); plot(fit2); par(mfrow=c(1,1))
```